Yuning Wu

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RESEARCH INTEREST

Design, build, and optimize AI-driven robots to humanely support humans in labor-intensive tasks.
Human-robot interaction, reinforcement learning, human-centered AI, robotics, computational design

EDUCATION

Carnegie Mellon University, Pittsburgh, USA
2024.5 (expected) **Ph.D. in Computational Design**
Dissertation Title: *Towards Humane Automation: An RL-Driven Robotic Framework for Contenxtually Supporting Labor-Intensive Construction Workers*
Co-Advisors: Daniel Cardoso Llach (Computational Design), Jean Oh (Robotics)

2021.5 **M.S. in Machine Learning**
Independent Research Advisor: Katerina Fragkiadaki

Tsinghua University, Beijing, China
2017.7 **M.Arch. & B.Arch in Architecture**

INDUSTRY EXPERIENCE

2021.5 - 2022.5 **Autodesk AI Lab & Robotics Lab**
Research Scientist Intern and Collaborator
- Work on Inverse Reinforcement Learning (IRL) methods for efficient robot learning from limited human demonstrations.
- Autodesk AI Lab has endowed a generous grant to my lab in my name after our work together, further strengthening the research collaboration.

2020.8 - 2023.12 **Carnegie Mellon University**, Pittsburgh, USA
Graduate Student Researcher & Technical Lead
- Lead the entire technical effort of Rethinking Automation in Construction.
- Design,prototype, and optimize a “work companion robot” for real-world challenges.
- Demonstrate and test the robot in a real construction site with workers.

2017.12 - 2019.8 **Metascope Technology**, Beijing, China
Co-founder and CTO
- Build a vision-based aesthetics recommendation system for helping commercial architects in forming client-specific design of space.
- Customize DNN models for recognizing visual architectural styles.
- Use statistical machine learning methods to design user profiling models.

RESEARCH PROJECTS & ROBOTS

Towards Humane Automation: An RL-Driven Robotic Framework for Supporting On-Site Construction Workers [1]
Husky is an RL-driven “work companion rover” built for robotically supported collaborative work. The robot can support and accompany on-site carpentry workers by adaptively navigating the complex, real-world construction site while comfortable delivering and carrying tools and materials around.
- Develop a novel pipeline to finetune and align RL models to context-specific worker preferences.
- Design the complete ROS-based, multi-sensor robotic prototype from a simple chassis.
- Use SoTA LiDAR Odometry and Mapping (LOAM) methods for unstructured site perception.
- Create an efficient multimodal hardware and software stack for perceiving workers’ activities and behavioral patterns.

A Distributed, Scalable RL-Driven Multi-Drone Framework for Supporting Flexible Tasks of Complex Structures [2]
Pixhawks is a fleet of drones driven by RL-based, multi–agent collision avoidance algorithms. The algorithmic framework enables a variable number of drones to adaptively execute simple construction tasks, such as pick-and-place and spray-coating tasks.
- Train and deploy a scalable, decentralized, RL-based multi-agent collision avoidance algorithm.
- Customize a generalizable Sim2Real pipeline using Unity MLAgents.
- Propose a proof-of-concept experiment with a physical drone prototype.
- Explore the system of integrating the drones into real-world work settings.
- Reflect on the evolving role of human workers in the emerging technical system.

Learning Efficient Dense Reward for RL-Based Robotic Manipulation from Human Demonstrations [3]
A plug-n-play dense reward for RL —a continuous metric that reflects task progress— is raised from multimodal sensory data for learning contact-rich manipulation tasks from a limited number of human demonstrations.

- Design a novel Temporal Variant Forward Sampling (TVFS) method that is more robust and cost-efficient in generating samples from human demonstrations.
- Raise a self-supervised latent representation learning architecture that can efficiently utilize sample pairs from TVFS.

MoonRanger for Discovering Water on the Moon
MoonRanger is an autonomous, suit-case-sized exploration rover searching for water and ice on the lunar surface. It was scheduled to be launched through the NASA CLPS project.

- Participate in system engineering and high-level organization of robot components.
- Create visualization and coordination dashboard for efficient progress updates and issue diagnosis.

Teaching AI Spatial Sense by Playing with LEGOs
LEGO Player is an AI agent that can stack stable structures using LEGO blocks based on a trained physics model.

- Build a GNN-based physics engine to infer the stability of a stacked LEGO structure.
- Use hierarchical reinforcement learning (HRL) to search for valid block stacking patterns.

RESEARCH PAPERS

[1] **Y. Wu**, J. Wei, J. Oh, D. Cardoso Llach. ***Towards Human-Centered Construction Robotics: An RL-Driven Companion Robot For Contextually Assisting Carpentry Workers***. Under review. Submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

[2] **Y. Wu***, E. Erdolu*, J. Wei, J. Oh, D. Cardoso Llach. ***Construction as Computer-Supported Cooperative Work: Designing a Robot for Assisting Carpentry Workers***. Under review. Submitted to ACM Conference on Computer Supported Cooperative Work (CSCW).

[3] Z. Fang, **Y. Wu**, A. Hassonjee, A. Bidgoli, D. Llach. ***Towards an Architectural Framework for Distributed, Robotically-Assisted Construction: Using Reinforcement Learning to Support Scalable Multi-Drone Construction in Dynamic Environments***. In Proceedings of the 40th Annual Conference of the Association for Computer-Aided Design in Architecture, ACADIA 2020.

[4] **Y. Wu**, J. Luo, H. Li. ***Learning Dense Rewards with Temporal Variant Self-Supervision***. International Conference on Robotics and Automation, ICRA 2022. Reinforcement Learning for Contact-Rich Manipulation Workshop.

[5] O. Akin, **Y. Wu**. ***State Space Paradox of Computational Research in Creativity***. In the 19th Heron Island Conference on Computational and Cognitive Models of Creative Design, HI 2019.

AWARDS AND HONORS

2023 & 2024	**CMU Presidential Fellowship**. Carnegie Mellon University, TCS. (Highest Honor in CMU)
2022	**Manufacturing Futures Initiative Fellowship**. Carnegie Mellon University.
2022	**Autodesk Research Merit Scholarship**. Autodesk AI Lab.
2021	**Manufacturing Pennsylvania Fellowship**. State of Pennsylvania.
2020	**GSA-Provost GuSH Grant**. Carnegie Mellon University.
2019	**LEGO Scholarship**. LEGO Foundation.
2017	**Distinctive Graduation Thesis**. Tsinghua University.
2011-2014	**Siyuan Exceptional Student Fellowship**. Tsinghua University.

INVITED TALKS & PRESENTATIONS

2023.5	**Invited Talk of PhD Research, Tata Innovation Forum 2023.** TCS, Cornell Tech.
2023.5	**Robot Presentation and Demo.** Bosch Leadership and AI Team.
2022.11	**CMU MFI Research Seminar Series**. CMU Manufacturing Futures Institute.
2022.10	**Situating Research**. CMU 48-620 Graduate Seminars.
2022.5	**Poster Presentation.** ICRA Reinforcement Learning for Contact-Rich Manipulation Workshop.
2021.11	**Inquiries into Machine Learning and Design**. CMU 48-770 Graduate Seminars.
2022.11 & 2021.11	**Robot Demonstration.** CMU MFI Annual Technical Exchange.
2020.10	**Poster and Presentation**. ACADIA "Distributed Proximities" Poster Sessions.

SKILLS FOR AI & ROBOTICS

AI/ML	**PyTorch / HuggingFace / Numpy / Tensorboard**
Simulation	**NVIDIA Omniverse Isaac Sim / Unity MLAgents / Mujoco / PyBullet**
Robotics	**ROS / ROS2 / OpenMMLab / OpenCV**
Coding	**Python / C++ / C#**
3D Modeling	**Rhinoceros / Grasshopper / Autodesk Series Softwares**